Unified Auto ML System for Classification & Regression

Automating Machine Learning Pipeline

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TOOLS USED : PANDAS, NUMPY, SCIKIT LEARN, STREAMLIT

Problem Statement

- Building Machine Learning Models often requires multiple manual process like preprocessing, model selection and fine tuning
- Where you have to feature scale and feature encode the data set
- Handling real-world data challenges like missing values and class imbalance adds complexity.
- Also choosing the right algorithm for a dataset requires multiple iterations and it is time consuming

Project Objective

- Develop a unified AutoML system that automates the end-to-end machine learning process.
- Automatically detect whether the problem is classification or regression.
- Apply preprocessing techniques like missing value handling and encoding and feature scaling.
- Train multiple models and perform hyperparameter tuning.
- Select the best-performing model based on evaluation metrics.
- Provide predictions and allow model download for future use.

Dataset Overview

- The system works with any CSV dataset uploaded by the user
- Two Examples of dataset used
- Income Data – Classification Problem
- Insurance Data - Regression Problem
- The system is designed to be dataset-independent and flexible.

Workflow

User Uploads CSV File

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Select Target Column

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Problem Type Detection (Classification / Regression)

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Data Preprocessing

- Missing Value Handling (Median / Mode)
- Encoding (One-Hot Encoding)
- Scaling (Standardization)

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Model Training

- Logistic Regression
- Random Forest
- XG Boost

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Hyperparameter Tuning (Randomized Search CV)

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Model Evaluation (Cross Validation)

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Best Model Selection

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Prediction + Feature Importance + Model Download

Key Techniques Used

- Pipeline : Combines preprocessing and model into a single workflow
- Handle missing value : Median for numerical, most frequent for categorical
- Feature Encoding: One-Hot Encoding for categorical variables
- Feature Scaling: StandardScaler for numerical features
- Cross Validation: Ensures model generalization
- Hyperparameter Tuning : RandomizedSearchCV used to find optimal parameters
- Auto Model Selection: Best model chosen based on performance

Models Used

- Logistic Regression :

  Used for classification problems

  Simple and interpretable
- Random Forest :

  Ensemble model using multiple decision trees

  Handles non-linearity and improves accuracy
- XGBoost :

  Advanced boosting algorithm

  Provides high performance and efficiency

Key Features

- Automatic problem type detection
- Handles missing values (Median / Mode)
- Handles class imbalance using class weights
- Supports both classification and regression
- Hyperparameter tuning using RandomizedSearchCV
- Feature importance visualization
- Model download for future use

Future Scope

- Add advanced evaluation metrics (ROC Curve, Confusion Matrix)
- Add an option to predict using user input
- Add an option to upload a new csv data and download the predictions
- Support more advanced models and deep learning

Conclusion

- Successfully developed a AutoML system
- Reduced manual effort in building ML models
- Automated preprocessing, training, and model selection
- Achieved reliable performance across different datasets

THANK YOU